

TIC 273026743

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 2.782727 d (ALIASED, see below)
-> corrected period: 5.565453 d
Epoch (t0) 2308.3427 BTJD
Duration 0.1200 d
Depth 2.141 % (21,409 ppm)
SNR 1982.1

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

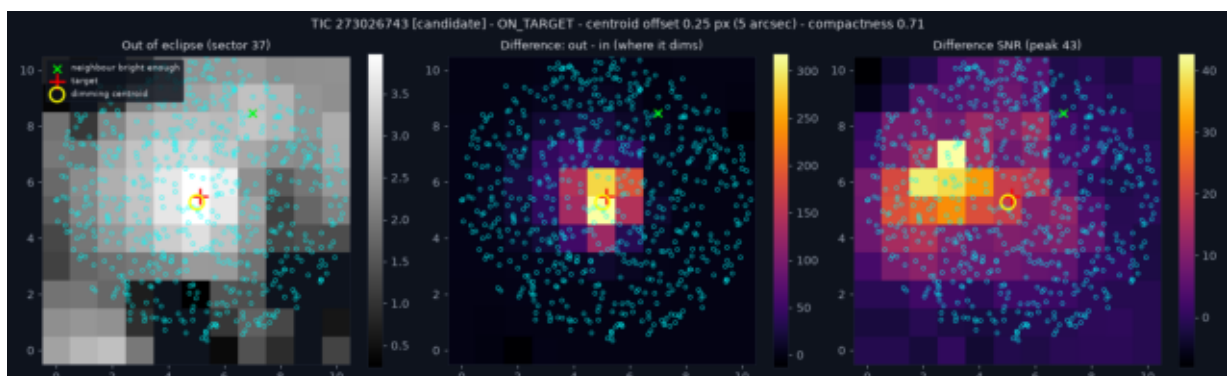
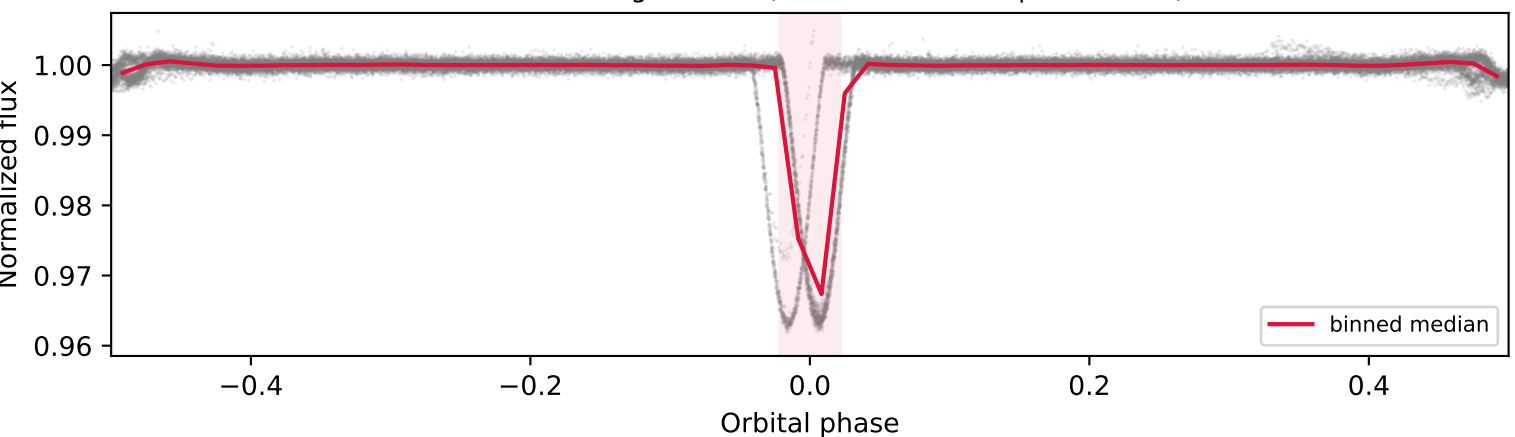
Eclipses observed 27 (gate ≥ 3)
Robust depth ratio 1.18 (gate ≥ 0.50)
Robust depth sigma 1566.7 (gate ≥ 7.0)
Odd/even z-score 21.38 (ALIAS FLAGGED)
Secondary eclipse detected, phase 0.49, 6.3 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 37 light-curve dip in this sector: 1061.1 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 42.77 (gate ≥ 5.0) Compactness 0.71 (gate ≥ 0.50)
Dimming centroid offset 0.25 px (5 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 273026689
3.74 px away, dTmag +2.34 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 21.38 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 5.565453 d, in any write-up.